

Question:

Every day, Wendi feeds each of her chickens three cups of mixed chicken feed, containing seeds, mealworms and vegetables to help keep them healthy. She gives the chickens their feed in three separate meals. In the morning, she gives her flock of chickens 15 cups of feed. In the afternoon, she gives her chickens another 25 cups of feed. How many cups of feed does she need to give her chickens in the final meal of the day if the size of Wendi's flock is 20 chickens?

Reasoning + Answer:

Let's break down the problem step by step .

1. **Understanding the daily feed requirement :**

- Wendi feeds each chicken 3 cups of feed per day .
- Wendi has 20 chickens .
- Therefore , the total daily feed requirement is :

$$\begin{aligned} & \left[\right. \\ & 20 \text{ chickens} \times 3 \text{ cups/chicken} = 60 \text{ cups} \\ & \left. \right] \end{aligned}$$

2. **Analyzing the morning and afternoon feed given :**

- In the morning , Wendi gives 15 cups of feed .
- In the afternoon , Wendi gives 25 cups of feed .
- Therefore , the total feed given in the morning and afternoon is :

$$\begin{aligned} & \left[\right. \\ & 15 \text{ cups} + 25 \text{ cups} = 40 \text{ cups} \\ & \left. \right] \end{aligned}$$

3. **Calculating the remaining feed needed :**

- The total daily feed requirement is 60 cups .
- The total feed given in the morning and afternoon is 40 cups .
- Therefore , the remaining feed needed in the evening is :

$$\begin{aligned} & \left[\right. \\ & 60 \text{ cups} - 40 \text{ cups} = 20 \text{ cups} \\ & \left. \right] \end{aligned}$$

<think>

- Wendi needs to give 20 more cups of feed in the evening .
- Each chicken gets 3 cups , and she has 20 chickens .
- The number of 3-cup portions in 20 cups is calculated as :

$$\begin{aligned} & \left[\right. \\ & \frac{20 \text{ cups}}{3 \text{ cups/chicken}} = \frac{20}{3} \approx 6.67 \text{ portions} \\ & \left. \right] \end{aligned}$$

- Since we can't split a chicken into fractions for practical purposes , we need to check the exact number of chickens :

$$\begin{aligned} & \left[\right. \\ & 20 \div 3 \approx 6 \text{ full portions and 2 additional cups , which is 1 more full chicken} \\ & \left. \right] \end{aligned}$$

- Thus , Wendi needs to give one more portion of 3 cups to complete the 20 cups .

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<answer>20 cups</answer>